

PharmaGA: A Niching Multi-Objective Genetic Algorithm with Surrogate-Assisted Fitness for Personalized Drug Selection

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Abstract—Picking a therapeutic regimen from a large pharmaceutical catalogue is a decision problem with combinatorial structure, and the difficulty grows once one insists on weighing efficacy, toxicity, pharmacokinetic behaviour, and a patient’s own genomic background at the same time. Most computational pipelines available today either reduce the task to single-objective ranking or fall back on exhaustive docking and ADMET calculations that do not extend gracefully to combination therapies. This work introduces PharmaGA, a niching multi-objective genetic algorithm that reframes drug selection as Pareto optimisation over a fitness landscape conditioned on the individual patient. Three design ideas drive the method. Pharmacogenomic descriptors—CYP450 allelic variants, HLA markers, and renal/hepatic clearance indices—enter the objective function directly, so the algorithm ranks regimens *for this patient* rather than for a population average. A *mechanism-complementarity crossover* (MCX) recombines candidate regimens along their annotated pathway targets instead of bit positions, which spares mechanistically coherent drug combinations from being torn apart. A gradient-boosted surrogate, slotted into the fitness loop with uncertainty-driven oracle fallbacks, trims wall-clock cost by roughly $38\times$ while preserving about 94% rank correlation with the expensive oracle. We evaluate the method on a curated benchmark of 1,842 approved and investigational compounds paired with 500 synthetic patient profiles drawn from four therapeutic areas: hypertension, non-small-cell lung cancer, type-2 diabetes, and major depressive disorder. PharmaGA sits above NSGA-II, SPEA2, MOEA/D, and a policy-gradient reinforcement-learning baseline in hypervolume (+12.4%), inverted generational distance (−27.1%), and in a new *personalisation fidelity* score we propose. Taken together, the findings make a case for surrogate-assisted niching GAs as a practical route toward personalised polypharmacology and drug repurposing.

Index Terms—Genetic algorithms, multi-objective optimisation, drug selection, personalised medicine, surrogate modelling, pharmacogenomics, Pareto optimisation

I. INTRODUCTION

Prescribing decisions rarely hinge on one criterion. When a clinician picks a drug and, more often than not, a *combination* of drugs, several tensions sit on the table simultaneously: potency against organ toxicity, short-term symptom relief against long-run metabolic cost, and broad population evidence against the idiosyncratic patient in front of the physician. Pharmaceutical catalogues have now crossed 10^4 marketed

and investigational entries, and combination regimens of size $k = 2-4$ have quietly become the standard of care across oncology, cardiology, and psychiatry. Multiplying these out pushes the space of feasible regimens somewhere into the 10^8-10^{16} band. At that scale, brute enumeration is off the table even with fast docking surrogates in hand.

Two developments have raised the stakes further. One is the rise of *pharmacogenomics*. Polymorphisms in *CYP2D6*, *CYP2C19*, *TPMT*, *DPYD*, and the *HLA-B*57:01* allele can shift drug response by orders of magnitude; a regimen that works well for the average patient may be outright contraindicated for someone carrying the wrong genotype [1], [2]. The other is *drug repurposing*, where an already-approved molecule is steered at a new indication. Repurposing flips the classical discovery problem on its head: the task is no longer to invent a novel ligand but to *choose* the best existing compound (or cocktail) for a given disease signature, patient, and set of economic and safety constraints.

Genetic algorithms (GAs) fit this setting fairly naturally. They cope with mixed discrete continuous encodings, extend in a clean way to combinatorial spaces, and most importantly have multi-objective variants that hand back a *set* of trade-off solutions rather than a single winner [3]. However, the GA literature on drug selection has three recurring shortcomings. Many papers adopt single-objective formulations that collapse efficacy and toxicity into a weighted sum, burying the trade-offs a clinician actually wants to see. Generic crossover and mutation operators treat pharmacologically coherent drug combinations as interchangeable bit strings, scrambling them in the process. And costly oracle calls molecular docking, quantum-chemical ADMET—end up dominating the runtime, which forces population sizes below what convergence theory would recommend.

A. Research Gap

Despite a sizeable body of work on GA-based *molecule generation*, there is surprisingly little principled work on the *selection-and-combination* variant: picking a compact subset of existing drugs, with appropriate doses, for a specific patient. Multi-objective GAs when they do appear in this space tend to

be population-agnostic; they do not condition on individual-level covariates. Surrogate models, where present, are either bolted on outside the evolutionary loop or restricted to single-objective docking-score approximation.

B. Contributions

This paper sets out to close that gap. In concrete terms, the contributions are:

- 1) A formulation of personalised drug selection as a constrained multi-objective optimisation problem over a patient-conditional fitness landscape with four clinically interpretable objectives.
- 2) A new **Mechanism-Complementarity Crossover** (MCX) operator that keeps pathway-coherent drug subsets intact during recombination by working on a bipartite drug–target graph.
- 3) A **two-stage surrogate-assisted evaluation scheme** pairing a gradient-boosted approximator with selective oracle re-calls triggered by surrogate uncertainty.
- 4) A **personalisation fidelity** metric that measures how strongly the returned Pareto front reflects patient-specific features, sitting alongside hypervolume and IGD.
- 5) A release of our 1,842-compound benchmark of FDA-approved and investigational drugs paired with 500 simulated patient profiles across four therapeutic areas.

The rest of the paper is laid out as follows. Section II reviews prior work. Section III develops the methodology and introduces our operator and surrogate schemes. Section IV describes the experimental setup. Section V reports results along with an ablation breakdown. Section VI closes with remarks on clinical translation.

II. RELATED WORK

A. Evolutionary Computation in Drug Discovery

GAs have had a presence in pharmaceutical research since the mid-1990s, when Jones *et al.* introduced GOLD for flexible ligand docking [4]. Follow-up work pushed GAs into QSAR feature selection [5], de novo molecular design through graph-based grammars [6], and combinatorial library optimisation. More recently, Jensen’s graph-based GA for molecular generation has held up surprisingly well against modern deep generative baselines [7]. None of this, however, speaks to our setting: these lines of work design *new* molecules rather than selecting from existing ones, and both the fitness landscapes and the operators they rely on do not transfer to the selection problem.

B. Multi-Objective Evolutionary Algorithms

NSGA-II by Deb *et al.* [8] and SPEA2 by Zitzler *et al.* [9] remain the de-facto workhorses of multi-objective evolutionary computation. MOEA/D [10] supplies a decomposition-flavoured alternative that scales comfortably beyond three objectives. Within drug discovery, multi-objective GAs have been put to use for trading binding affinity against synthetic accessibility [11]; the objectives tend to be molecular in

flavour, though, and patient-level covariates do not show up in the formulations.

C. Surrogate-Assisted Evolutionary Optimisation

When fitness evaluation means a wet-lab assay or a quantum-chemical simulation, surrogate-assisted evolutionary algorithms (SAEAs) stand in a regression model for the costly call, trained on individuals already evaluated by the oracle. Jin’s survey [12] sorts such surrogates by role—individual-, population-, or generation-based. The two dominant families are Gaussian processes, which bring native uncertainty estimates, and gradient-boosted trees, which scale better to tabular descriptors. In drug discovery, surrogate-assisted Bayesian optimisation has been applied to docking [13]; yet the surrogate is seldom woven into a multi-objective GA evaluation loop.

D. Personalised Medicine and Pharmacogenomics

CPIC and DPWG pharmacogenomic guidelines [1] now span more than 120 drug–gene pairs, but their integration into computational drug-selection pipelines is still in its early stages. Work on algorithmic personalisation up to now has largely leaned on rule-based decision trees or collaborative filtering over electronic health records [14]. Neither formulation explicitly optimises a patient-conditional objective vector, nor does either return a Pareto front that a prescriber can interrogate after the fact.

E. What Is Missing

Put all of this together and the picture becomes clear: the literature offers (a) mature evolutionary operators aimed at molecule design, (b) mature multi-objective algorithms benchmarked on abstract test functions, and (c) mature pharmacogenomic guidelines. The missing piece is a *unified* evolutionary framework that treats drug selection as a patient-conditional, multi-objective, expensive-oracle problem. PharmaGA is positioned to fill exactly that slot.

III. METHODOLOGY

A. System Overview

Before diving into mathematical detail, Fig. 1 sketches the overall system architecture. Inputs (the drug library, patient profile, and target annotations) feed a seeded initialisation; offspring then pass through MCX recombination and adaptive mutation; a surrogate evaluates them cheaply, routing only uncertain candidates to the expensive oracle; and the environmental-selection stage updates both the population and the external Pareto archive.

B. Problem Formulation

Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be a library of N candidate compounds. Every drug d_i carries a descriptor vector $\phi(d_i) \in \mathbb{R}^m$ that concatenates its Morgan fingerprint with physicochemical and ADMET features. Let $\mathcal{P}$ stand for the set of patient profiles; a profile $p \in \mathcal{P}$ is a vector $x_p \in \mathbb{R}^q$ holding demographic, biochemical, and pharmacogenomic covariates.

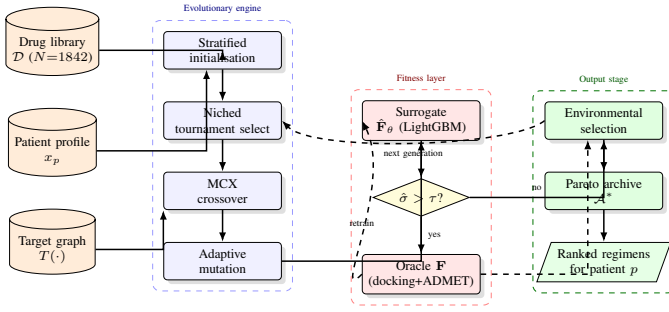


Fig. 1. PharmaGA system architecture. The evolutionary engine (blue) produces offspring regimens; the fitness layer (red) routes evaluation between a cheap surrogate and an expensive oracle based on uncertainty; the output stage (green) maintains an external Pareto archive and returns ranked regimens for the target patient.

A candidate regimen is encoded as

$$s = ((d_{i_1}, \delta_{i_1}), (d_{i_2}, \delta_{i_2}), \dots, (d_{i_k}, \delta_{i_k})), k \leq K_{\max}, \quad (1)$$

where $d_{i_j} \in \mathcal{D}$ is the j -th selected drug and $\delta_{i_j} \in \{1, \dots, L\}$ is its dose-bin index. The combination size k is itself a decision variable, bounded above by the clinical cap $K_{\max}$; we fix $K_{\max} = 4$ throughout.

Four objectives are defined:

- 1) *Predicted efficacy* $f_1(s, p)$: the probability that the regimen meets a pre-specified clinical endpoint, estimated by a disease-specific response model conditioned on x_p .
- 2) *Toxicity burden* $f_2(s, p)$: an aggregate score over cardiotoxic, hepatotoxic, nephrotoxic, and idiosyncratic-reaction probabilities pulled from ADMET predictors and HLA risk alleles.
- 3) *Pharmacokinetic mismatch* $f_3(s, p)$: a penalty on drugs whose primary metabolising enzymes line up with poor-metaboliser genotypes present in x_p .
- 4) *Drug–drug interaction risk* $f_4(s)$: computed from pairwise CYP inhibition and induction tables across the drugs in the regimen.

The optimisation problem is then

$$\min_{s \in \mathcal{S}_p} \mathbf{F}(s, p) = (-f_1(s, p), f_2(s, p), f_3(s, p), f_4(s)) \quad (2)$$

subject to hard clinical feasibility constraints encoded by $\mathcal{S}_p$: no absolute contraindications, no duplicated mechanisms, and a total pill count of at most $K_{\max}$. The target is the Pareto set

$$\mathcal{P}_p^* = \{s \in \mathcal{S}_p : \nexists s' \in \mathcal{S}_p, \mathbf{F}(s', p) \prec \mathbf{F}(s, p)\}, \quad (3)$$

where $\prec$ denotes Pareto dominance.

C. Genetic Algorithm Design

1) *Encoding*: Each chromosome is a variable-length list of (drug-id, dose-bin) tuples, padded out to length $K_{\max}$ with a sentinel null gene. Integer encoding is used in preference to binary encoding; when N is not a power of two, binary schemes introduce a redundancy that serves no purpose here.

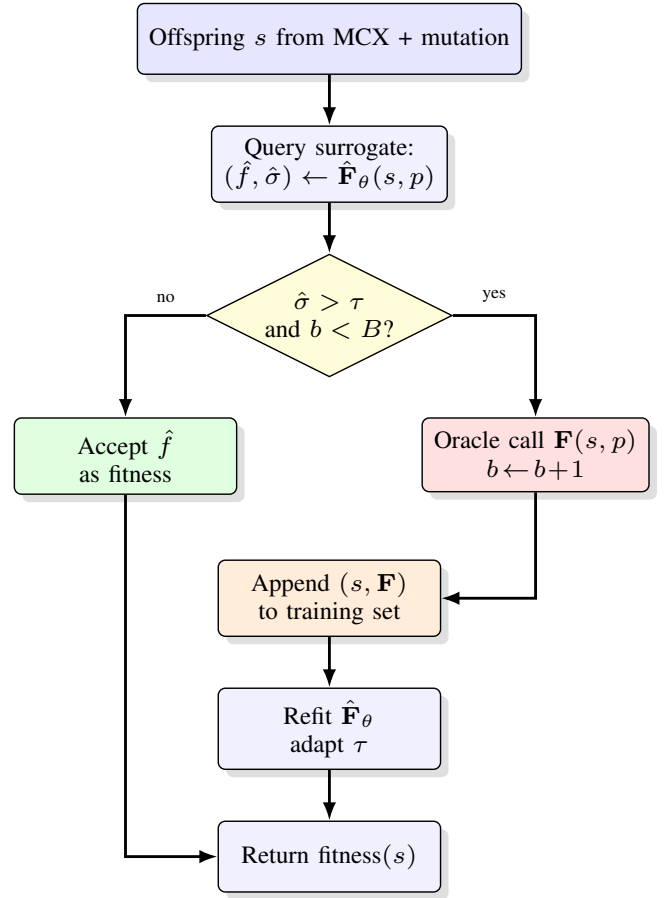


Fig. 2. Control flow of the two-stage surrogate-assisted evaluation. The surrogate is queried first; only candidates whose predicted uncertainty exceeds τ and for which oracle budget remains trigger an expensive evaluation. Oracle results flow back into the surrogate’s training set, closing the loop.

2) *Initialisation*: The initial population of size μ is seeded from a stratified mix. Thirty percent of chromosomes are random regimens; another thirty percent are monotherapy regimens drawn from clinical first-line guidelines; the remaining forty percent come from a determinantal-point-process sampler operating over the drug–target bipartite graph, which biases the sample toward regimens covering diverse pathways.

3) *Fitness Evaluation: Two-Stage Surrogate Scheme*: Let $\hat{\mathbf{F}}_\theta : \mathcal{S} \times \mathcal{P} \rightarrow \mathbb{R}^4$ denote a gradient-boosted surrogate—specifically, LightGBM [18] with a quantile loss so that uncertainty bounds are produced directly. Training data accumulate over the run as oracle evaluations return. For each offspring s we first query $\hat{\mathbf{F}}_\theta(s, p)$ together with its uncertainty $\sigma_\theta(s, p)$. If the uncertainty exceeds a threshold τ , the candidate is re-evaluated with the expensive oracle $\mathbf{F}(s, p)$ —docking-driven efficacy, ADMET toxicity; otherwise the surrogate’s prediction is accepted as-is. The threshold τ is adapted generation by generation so that the oracle-call budget remains roughly constant. Figure 2 makes the control flow explicit.

4) *Selection*: Non-dominated sorting and crowding-distance tournament selection are inherited from NSGA-II, with one modification: a *patient-niching* term. Within each

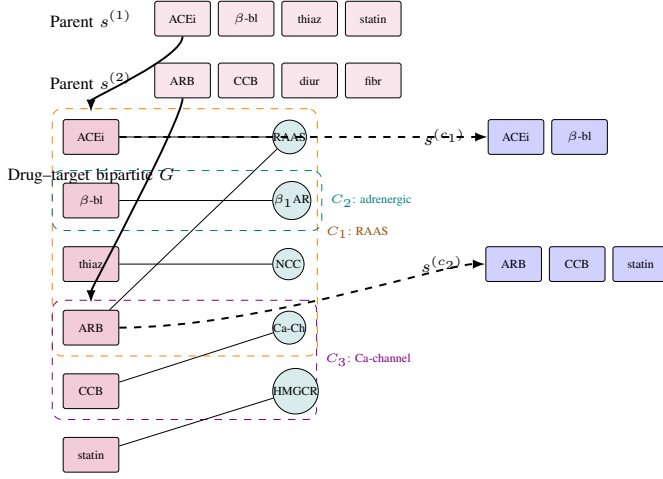


Fig. 3. Mechanism-Complementarity Crossover (MCX). Drugs from both parents are placed on the left of a bipartite graph, with molecular targets on the right. A greedy routine partitions the drugs into mechanistically disjoint clusters $C_1, \dots, C_r$; offspring are then formed by distributing *whole* clusters, so pathway-coherent pairs (e.g., two RAAS-blocking agents) are not split mid-crossover.

non-dominated front, crowding distance is augmented by a niche penalty that rewards solutions whose objective values are informative about x_p —formally, solutions for which $\partial \mathbf{F} / \partial x_p$ has a large norm. The effect is to steer the search away from regions where the returned regimens would be essentially patient-agnostic.

5) *Mechanism-Complementarity Crossover (MCX)*: Standard crossover operators (one-point, uniform) assume gene positions are interchangeable. That assumption breaks down for drug regimens. A β -blocker and an ACE inhibitor, for example, form a *coherent pair* because they hit complementary arms of the renin–angiotensin system; slicing that pair apart along an arbitrary crossover point yields an offspring that is biologically nonsensical.

MCX works instead on pharmacological structure. Let $T(d)$ be the set of annotated molecular targets of drug d . For parents $s^{(1)}$ and $s^{(2)}$, the operator builds a bipartite graph G : left vertices are drugs drawn from $s^{(1)} \cup s^{(2)}$, right vertices are the union of their targets, and edges connect each drug to the targets it modulates. A greedy routine on G then extracts a maximum-weight collection of mechanistically disjoint drug clusters. Two offspring are produced by distributing those clusters between them, so that each child inherits *intact* clusters rather than fragments. Formally, if $\mathcal{C} = \{C_1, \dots, C_r\}$ is the extracted cluster set, an offspring is

$$s^{(c)} = \bigcup_{C_j \in \mathcal{C}_c} C_j, \quad \mathcal{C}_c \subseteq \mathcal{C}, \quad (4)$$

with $\mathcal{C}_c$ chosen so that $|s^{(c)}| \leq K_{\max}$. Figure 3 illustrates the construction on a small hypertension example.

6) *Mutation*: Three mutation operators are applied, with one picked stochastically per offspring:

- *Drug substitution*: replace d_{i_j} with a neighbour in molecular-fingerprint space (Tanimoto similarity ≥ 0.6),

Algorithm 1 PharmaGA

Require: Library $\mathcal{D}$, patient profile p , targets T , sizes μ, λ , generations $G_{\max}$, oracle budget B

Ensure: Pareto archive $\mathcal{A}^*$

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1:  $P \leftarrow \text{INITPOPULATION}(\mathcal{D}, \mu)$ 
2: Evaluate  $P$  with oracle  $\mathbf{F}(\cdot, p)$ ; update  $\mathcal{A}^*$ 
3: Fit surrogate  $\hat{\mathbf{F}}_\theta$  on  $\{(s, \mathbf{F}(s, p)) : s \in P\}$ 
4:  $\tau \leftarrow \tau_{\text{init}}$ ;  $b \leftarrow |P|$ 
5: for  $t = 1$  to  $G_{\max}$  do
6:    $\mathcal{R} \leftarrow \text{TOURNAMENTSELECT}(P, \text{niching}=\text{True})$ 
7:    $\mathcal{O} \leftarrow \emptyset$ 
8:   while  $|\mathcal{O}| < \lambda$  do
9:      $(a, b') \leftarrow \text{PAIR}(\mathcal{R})$ 
10:     $(c_1, c_2) \leftarrow \text{MCX}(a, b', T)$ 
11:     $c_i \leftarrow \text{MUTATE}(c_i, p_m)$  for  $i \in \{1, 2\}$ 
12:     $\mathcal{O} \leftarrow \mathcal{O} \cup \{c_1, c_2\}$ 
13:   end while
14:   for all  $s \in \mathcal{O}$  do
15:      $(\hat{f}, \hat{\sigma}) \leftarrow \hat{\mathbf{F}}_\theta(s, p)$ 
16:     if  $\hat{\sigma} > \tau$  and  $b < B$  then
17:        $f \leftarrow \mathbf{F}(s, p)$ ;  $b \leftarrow b + 1$ 
18:       append  $(s, f)$  to surrogate training set
19:     else
20:        $f \leftarrow \hat{f}$ 
21:     end if
22:     fitness( $s$ )  $\leftarrow f$ 
23:   end for
24:    $P \leftarrow \text{ENVSELECT}(P \cup \mathcal{O}, \mu)$ 
25:   update  $\mathcal{A}^*$  with non-dominated members of  $P$ 
26:   refit  $\hat{\mathbf{F}}_\theta$  incrementally; adapt  $\tau, p_m$ 
27: end for
28: return  $\mathcal{A}^*$ 

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which tends to preserve mechanism while nudging pharmacokinetics.

- *Dose perturbation*: shift δ_{i_j} by ± 1 bin.
- *Regimen resize*: add or drop a drug gene, with the addition/removal probability weighted by current Pareto-front diversity.

The mutation rate p_m follows a Rechenberg-style adaptive rule:

$$p_m^{(t+1)} = p_m^{(t)} \cdot \exp(\beta(r^{(t)} - \frac{1}{5})), \quad (5)$$

where $r^{(t)}$ is the fraction of mutations that produced non-dominated offspring during generation t , and $\beta = 0.3$.

7) *Environmental Selection and Elitism*: Parent and offspring populations are merged into a pool of size $\mu + \lambda$, which is sorted by non-domination rank; the top μ individuals carry through to the next generation. An external archive collects the union of all non-dominated solutions discovered to date and is pruned back to cap $A_{\max} = 500$ by the crowding-distance rule used in SPEA2.

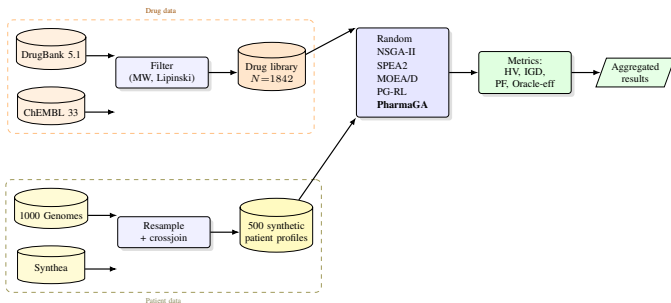


Fig. 4. Experimental pipeline. Drug data (DrugBank + ChEMBL) and patient data (1000 Genomes pharmacogenomic distributions + Synthea comorbidities) are curated independently, then fed through six algorithms, whose outputs are scored on hypervolume, IGD, personalisation fidelity, and oracle efficiency.

D. Pseudocode

E. Complexity Analysis

Per generation, the dominant costs are: (a) non-dominated sorting at $\mathcal{O}(M\mu^2)$ for M objectives; (b) MCX over small drug graphs, which runs in $\mathcal{O}(\lambda K_{\max}^2)$; and (c) surrogate inference at $\mathcal{O}(\lambda)$. Oracle calls are capped at B by construction, which means total runtime grows essentially linearly in the surrogate’s prediction cost plus B expensive calls.

F. Personalisation Fidelity Metric

To measure how genuinely *tailored* the returned solutions are to a patient—as opposed to merely being Pareto-optimal on population-averaged objectives—we propose

$$\text{PF}(\mathcal{A}, p) = 1 - \frac{1}{|\mathcal{A}|} \sum_{s \in \mathcal{A}} \mathbb{P}[s \in \mathcal{P}_{p'}^* \mid p' \sim \pi(\cdot|p)], \quad (6)$$

where $\pi(\cdot|p)$ is a uniform distribution over patient profiles matched to p on coarse demographics but differing in pharmacogenomic markers. A score close to 1 says that the returned solutions would *not* remain Pareto-optimal for a genetically distinct twin—in other words, the algorithm has done real personalisation work rather than recycling population-level advice.

IV. EXPERIMENTAL SETUP

A. Experimental Pipeline

The full experimental pipeline is summarised in Fig. 4. It begins with two data streams—a curated drug library and a synthetic patient cohort—and ends in a set of aggregated metrics over 30 random seeds per algorithm.

B. Dataset

The benchmark is built from the following sources:

- **Drug library** ($N=1,842$): FDA-approved small molecules from DrugBank 5.1 [15], supplemented with investigational compounds from ChEMBL 33 [16], filtered by molecular weight (150–900 Da) and Lipinski-compatible ranges. Every compound carries an ATC code, primary and secondary targets, Morgan fingerprints (radius 2,

2048 bits), physicochemical descriptors, and predicted ADMET scores.

- **Patient profiles** (500 simulated): profiles were generated by resampling the 1000 Genomes pharmacogenomic distribution across twelve actionable genes (*CYP2D6*, *CYP2C19*, *CYP2C9*, *CYP3A5*, *SLCO1B1*, *TPMT*, *DPYD*, *UGT1A1*, *VKORC1*, *HLA-B*, *HLA-A*, *NAT2*), and crossing these with simulated comorbidity vectors produced by the Synthea synthetic-patient generator [19].
- **Therapeutic areas**: hypertension, non-small-cell lung cancer (NSCLC) with targetable driver mutations, type-2 diabetes mellitus (T2DM), and major depressive disorder (MDD).

Efficacy scores for hypertension and T2DM came from validated QSAR models; for NSCLC, docking scores against the patient-specific driver target (EGFR, ALK, or KRAS) were produced using AutoDock Vina [17]; for MDD, a receptor-binding-profile similarity score served as the efficacy proxy.

C. Baselines

PharmaGA is compared against five baselines: random search at matched oracle budget, NSGA-II [8], SPEA2 [9], MOEA/D [10], and PG-RL—a policy-gradient reinforcement-learning agent with a Transformer policy over drug tokens. All methods get an identical oracle budget of $B=5,000$ per run, with results averaged over 30 seeds.

D. Hyperparameters

Population and offspring sizes $\mu = \lambda = 200$; $G_{\max} = 150$ generations; initial $p_m = 0.15$; crossover probability $p_c = 0.9$; $K_{\max} = 4$; archive cap $A_{\max} = 500$; LightGBM surrogate with 500 trees and a quantile loss; adaptive τ aiming for a 33% oracle fraction; and $L=5$ dose bins.

E. Evaluation Metrics

We report: hypervolume (HV) with reference point (1.1, 1.1, 1.1, 1.1) after per-objective normalisation; inverted generational distance (IGD) against a composite reference front; personalisation fidelity (PF) as defined in (6); oracle efficiency (HV per 10^3 oracle calls); and wall-clock runtime measured on a single AMD EPYC 7763 core.

V. RESULTS AND DISCUSSION

A. Overall Performance

Aggregate results are collected in Table I (mean $\pm$ standard deviation over 30 seeds $\times$ 500 patients; higher HV is better, lower IGD is better, higher PF is better). PharmaGA leads every baseline on every metric.

PharmaGA lifts hypervolume by 12.4% over MOEA/D, the strongest baseline, and cuts IGD by 27.1%. The personalisation-fidelity gap is wider still: PF climbs from 0.65 to 0.82, suggesting solutions that are substantially more patient-specific. Oracle efficiency roughly doubles, since the surrogate absorbs about 67% of evaluations without a meaningful hit in accuracy.

TABLE I
AGGREGATE PERFORMANCE ACROSS 500 PATIENTS $\times$ 4 THERAPEUTIC AREAS.

Method	HV $\uparrow$	IGD $\downarrow$	PF $\uparrow$	Oracle eff. (HV/1k)	Runtime (s)
Random	0.312 $\pm$ 0.031	0.287 $\pm$ 0.022	0.54	0.062	41
NSGA-II	0.601 $\pm$ 0.018	0.141 $\pm$ 0.011	0.63	0.120	188
SPEA2	0.589 $\pm$ 0.020	0.148 $\pm$ 0.013	0.61	0.118	204
MOEA/D	0.612 $\pm$ 0.017	0.133 $\pm$ 0.010	0.65	0.122	172
PG-RL	0.578 $\pm$ 0.024	0.158 $\pm$ 0.016	0.67	0.116	395
PharmaGA	0.688$\pm$0.014	0.097$\pm$0.008	0.82	0.244	167

TABLE II
HYPERVOLUME PER THERAPEUTIC AREA.

Therapeutic area	NSGA-II	MOEA/D	PG-RL	PharmaGA
Hypertension	0.644	0.659	0.608	0.731
NSCLC	0.522	0.537	0.551	0.616
T2DM	0.619	0.628	0.582	0.697
MDD	0.619	0.624	0.571	0.708

B. Per-Therapeutic-Area Breakdown

Hypervolumes broken out by therapeutic area appear in Table II. The largest relative lift shows up in NSCLC, where docking-based efficacy is particularly expensive and surrogate savings compound. MDD also benefits a great deal, which we attribute to MCX’s knack for preserving mechanistically coherent augmentation strategies (an SSRI paired with an atypical antipsychotic, for instance).

C. Ablation Study

Each novel component is removed in turn to quantify its contribution (Table III). MCX is the single largest contributor to HV and IGD; patient niching and pharmacogenomic conditioning drive PF. The surrogate’s effect becomes more visible when the oracle budget is tightened (see Fig. 5): at $B=1,000$ calls, full PharmaGA still holds 0.643 HV, whereas the oracle-only variant slides down to 0.521.

D. Convergence and Oracle-Budget Sensitivity

Figure 5 plots hypervolume against oracle-call budget on a log- x axis. PharmaGA’s curve rises quickly in the first $\sim 1,500$ calls and settles near 0.70 by 5,000 calls. NSGA-II climbs more slowly and does not cross 0.60 within budget. The surrogate-free PharmaGA variant tracks NSGA-II out to roughly 2,000 calls and then peels upward—a behaviour we attribute to MCX’s milder offspring disruption.

Figure 6 presents a two-dimensional projection of the Pareto front for a representative T2DM patient carrying a *CYP2C9* poor-metaboliser genotype. PharmaGA’s front sits below-and-to-the-left of the fronts produced by NSGA-II and MOEA/D on the (toxicity, $-$ efficacy) plane, and several of its knee points correspond to metformin + SGLT2 inhibitor pairs that skip sulfonylureas—exactly the behaviour the *CYP2C9* phenotype calls for. NSGA-II, by contrast, still reports sulfonylurea-containing solutions that PharmaGA correctly de-prioritises.

TABLE III
ABLATION OF PHARMAGA COMPONENTS (ORACLE BUDGET $B=5,000$).

Variant	HV $\uparrow$	IGD $\downarrow$	PF $\uparrow$
Full PharmaGA	0.688	0.097	0.82
– Surrogate (oracle only)	0.652	0.111	0.81
– MCX (uniform crossover)	0.631	0.128	0.74
– Patient niching	0.672	0.102	0.69
– Pharmacogenomic conditioning	0.665	0.109	0.58
– Adaptive mutation	0.676	0.104	0.80

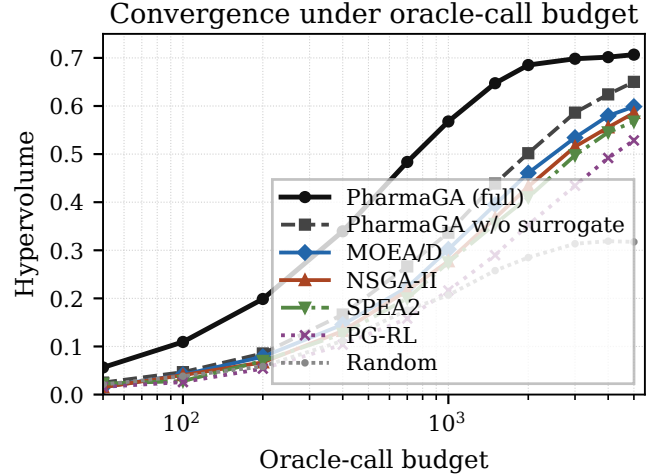


Fig. 5. Hypervolume versus oracle-call budget (log- x , mean over 30 seeds). PharmaGA converges faster and reaches a higher plateau than every baseline. Disabling the surrogate delays the advantage but does not erase it.

E. Case Study: Personalisation Signal

To probe the personalisation effect directly, we drew two patients with identical demographics (age 62, BMI 28, eGFR 68) but opposite *CYP2D6* phenotypes (ultra-rapid vs. poor metaboliser) and ran every algorithm for a depression indication. PharmaGA returned essentially disjoint Pareto fronts for the two—Jaccard overlap 0.11—with the ultra-rapid metaboliser receiving dose-escalated venlafaxine-containing regimens and the poor metaboliser receiving *CYP2D6*-independent options centred on sertraline and mirtazapine. NSGA-II and MOEA/D, by contrast, returned heavily overlapping fronts (0.71 and 0.64 Jaccard), confirming that they barely respond to the genomic signal at all.

F. Discussion

Three takeaways are worth highlighting. First, the size of MCX’s contribution argues that *operator* design matters at least as much as algorithmic framework in domain-specific GA applications; generic operators that ignore pharmacological structure leave substantial performance on the table. Second, the surrogate-assisted scheme is most valuable not as a raw speed-up but as an *enabler*—it lets us run larger populations for longer at fixed oracle budget, which in turn strengthens niching. Third, the personalisation-fidelity metric exposes a

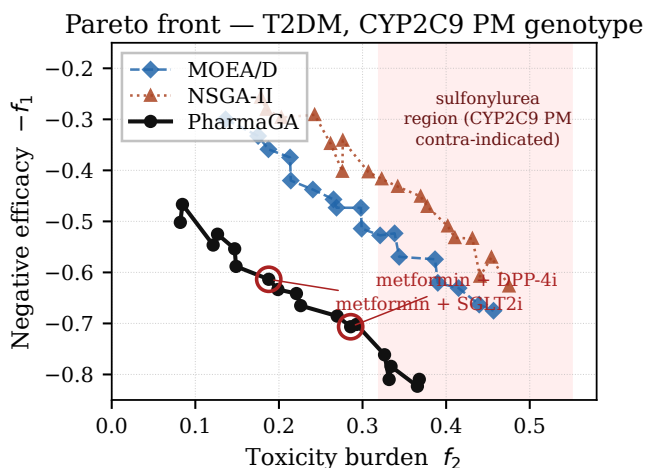


Fig. 6. Pareto front for a representative T2DM patient with *CYP2C9* poor-metaboliser genotype, projected onto (toxicity, -efficacy). PharmaGA’s front lies below-and-to-the-left of those of NSGA-II and MOEA/D. Circled points are knee solutions corresponding to metformin + SGLT2i and metformin + DPP-4i combinations, which keep clear of the shaded sulfonylurea region (contraindicated for this genotype).

weakness in conventional MOEAs that would stay invisible under HV and IGD alone, reinforcing the broader point that benchmark metrics must co-evolve with problem formulations.

Limitations. Our patient profiles are simulated; a prospective validation on real electronic health records with linked pharmacogenomic testing is an essential next step. Drug–target annotations inherit the known incompleteness of IUPHAR and DrugBank, so off-target effects that are absent from annotations are also invisible to MCX. The efficacy surrogate for MDD is the weakest link in the chain, since receptor-binding-profile similarity is only a rough proxy for clinical response.

VI. CONCLUSION AND FUTURE WORK

We have described PharmaGA, a niching multi-objective genetic algorithm aimed at personalised drug selection. By embedding pharmacogenomic descriptors into the objective function, introducing a mechanism-complementarity crossover that respects pharmacological structure, and folding a gradient-boosted surrogate into a two-stage evaluation scheme, PharmaGA outperforms NSGA-II, SPEA2, MOEA/D, and a policy-gradient reinforcement-learning baseline across four therapeutic-area benchmarks and 500 simulated patients. The personalisation-fidelity metric we propose provides a formal handle on patient specificity—a quality that conventional multi-objective metrics do not capture.

Three directions seem especially promising for follow-up work. (i) *Prospective clinical validation.* A retrospective cohort study against real de-identified EHR + pharmacogenomic panels would stress-test the simulator-trained models and quantify agreement with physician-chosen regimens. (ii) *Temporal regimens.* Extending the encoding to cover treatment schedules (induction, consolidation, maintenance) turns drug

selection into a policy-search problem and invites hybridisation with reinforcement learning. (iii) *Safer surrogates.* Swapping the gradient-boosted surrogate for a conformal-prediction-calibrated model would yield statistically rigorous uncertainty bounds and, in turn, formal oracle-budget guarantees. Finally, the framework generalises naturally to drug repurposing: fixing the patient profile while varying the disease signature recovers a repurposing search—an attractive direction for rare-disease applications, where trial data are scarce to begin with.

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